



Incorporating operational constraints into long-term energy planning: The case of the Egyptian power system under high share of renewables

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ABSTRACT

This study assesses the performance of Egypt's energy system on a short-term basis to ensure that it can handle a high share of renewable energy, as predicted in the long-term planning up to 2040. PLEXOS software is used for short-term analysis to simulate operational performance on an hourly basis while considering the constraints of Unit Commitment (UC) and Economic Dispatch (ED). The long-term energy planning model, which covered the years 2020–2040 with chosen representative years (2020 as the base year and 2025, 2030, 2035, and 2040), was validated by the short-term operational model. The results from the short-term analysis indicate that by the year 2040, 69.5 % of the electricity mix from renewable energy sources (excluding hydro) can be accommodated, representing a slightly lower capacity than the 70 % predicted by the long-term plan. Due to the high share of intermittent renewable energy sources, traditional generators will experience excessive cycling in 2040. The average weighted short-run marginal cost in 2040 is projected to be 7.8 USD/MWh, compared to 18.37 USD/MWh in 2020, due to the low operating costs of renewable energy plants and their high penetration in 2040. Various operational techniques have been suggested to reduce the curtailment of renewable energy to achieve only 1.2 % of the total energy generation of the system by 2040. Additionally, other adjustments to the system operation are being considered to maintain a balance in the power system on a short-term basis.

Abbreviations:

AEM	Accounting energy models
CAPEX	Capital expenditure
CCGT	Combined-cycle gas turbine
CCPP	Combined-cycle power plant
CSP	Concentrated solar power
ED	Economic dispatch
EEHC	Egyptian Electricity Holding Company
GDP	Gross Domestic Product
HEM	Hybrid energy models
LCOE	Levelized Cost of Energy
MIP	Mixed integer programming
NPP	Nuclear power plant
NREA	New & Renewable Energy Authority, Egypt
OCGT	Open-cycle gas turbine
OEM	Optimization energy models
O&M	Operating and maintenance cost
PHS	Pumped hydro storage
PV	Photovoltaic
RE	Renewable energy
SEM	Simulation energy models

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SPP-G	Gas-fired steam power plant
SRMC	Short-run marginal cost
UC	Unit commitment
VOM	Variable operation and maintenance
VRE	Variable renewable energy

1. Introduction

The existing global energy system is almost reliant on hydrocarbons [1]. In the coming decades, global energy markets will confront numerous challenges. These challenges include decarbonization, power systems infrastructure restructuring, support for renewable technologies, and preservation of supply reliability and security. To tackle these challenges, governments around the world are establishing targets and strategies to combat climate change and global warming [2]. These targets and strategies should include low-emission energy resources such as wind [3] and solar [4]. The anticipated deployment of renewable

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electricity generation marks a significant change in the operation of the electricity system due to the intermittent nature of these resources. The stochastic nature of these resources makes renewable energy forecasting a complex process, which may create operational challenges for the thermal power plants by necessitating recurring cycling, including ramping up/down and startup/shutdown [5]. To integrate high levels of variable renewable energy (VRE) into power systems, future generation portfolios must be flexible enough to deal with frequent and abrupt changes in the generation mix. In addition, maintaining system adequacy and reliability is required [6]. This necessitates analyzing the power system in the short term (on hourly or even less time steps) to discover challenges for integrating renewable energy systems with the power system [7].

Long-term energy models are used to aid policymakers and strategic planners in understanding the future expansion complexity of energy systems. These models can simulate the complete energy system or specific components for decades ranging from 20 to 50 years in the future [8]. These models are facing challenges such as combining environmental policies, the restructuring of the energy sector (privatization, etc.), energy policy goals (energy security and affordable energy supply) and energy efficiency targets, use of indigenous and renewable energy sources, ...etc. Long-term energy models provide quantitative tools to analyze the different options to assist policymakers in making better decisions. To address all these issues, an integrated energy planning model should be used, which includes a representation of the whole energy system and its environmental impacts. The energy policy models can be classified into four categories. These include Optimization energy models (OEM), Simulation energy models (SEM), Accounting energy models (AEM), and Hybrid energy models (HEM). Regarding the OEM, they are based on linear programming to identify the least cost of an energy system to satisfy a specified demand based on a set of constraints (e.g., supply constraints, technology availability, environmental constraints, ...etc.). This is achieved by selecting different supply technologies according to their lifecycle costs. The OEMs are data-intensive models. They assume perfect competition, such that prices are the only factor in defining the share of each supply technology. They do not take into consideration the policy options, which are not technology-based. MARKAL/TIMES, EFOM, and WASP are examples of these Optimization Models [9]. The SEM simulates the behavior of energy consumers and producers under various market signals (e.g., price, income levels, limits on the stock turnover rate). They do not assume perfect competition in the market, such that the share of a technology is not solely dependent on price consideration. For example, a technology share can depend on both the price and quality of energy services. However, they are data-intensive models. Furthermore, projections are sensitive to the starting conditions and parameters, which are sometimes hard to define accurately. Examples of this category of energy models are ENPEP and Energy 20/20 [10]. Regarding AEM, they rely on developing and analyzing various scenarios by contrasting their energy needs, social costs and benefits, and environmental implications. They do not identify technological market share based on prices or other criteria. This enables decision-makers to evaluate both the effects of a single policy and the interactions resulting from combining several policies and measures. They enjoy the advantage of having simple and lower data requirements. Furthermore, they do not assume perfect competition. They can be used to explore options that go beyond cost or are sometimes not easy to cost. Alternatively, they do not identify the least cost option. Furthermore, they are less suitable where systems are complex. They are suitable for developing countries, where data may be contingent, and the systems are more straightforward. This energy model category includes LEAP, MEDEE, and MESAP [11]. A study by Malka et al. [12] used a LEAP model customized for Norway to propose mitigation scenarios to help reach environmental goals by 2050. Regarding the HEM, they combine elements of optimization, simulation, and accounting models to avoid the shortcomings of each category. They may take the form of a two-level model, where the lower level typically follows the

accounting scheme, while the top follows the optimization scheme. An alternate strategy is to implement a general equilibrium system that combines a simulation method for each demand sector with an optimization module for the electricity sector. TIMES/Egypt, a customized version of Markel/TIMES, has adopted this approach. It is a scenarios-based model with an optimization top to identify the least cost solution within each scenario. TIMES/Egypt has been used for the long-term planning of Egypt's energy system. Since the long-term planning models cover a wide range of projected years, which can extend to 50 years, they have a low resolution. According to Ludig et al. [13], having a too-coarse depiction of variability can lead to overestimating the total system cost. Loisel et al. [14] presented a soft coupling between the long- and short-term planning of the French power system, where PLOES energy model is used for the long-term. Validating the long-term on a short-term basis is done on EcoNUK, a power market model applied to France. Results show that the French power system in 2050 could be balanced with 26 % nuclear, 71 % renewables, and the rest is gas plants for balancing the grid with energy storage.

The depiction of short-term variability becomes crucial as the amount of intermittent renewable energy in the power system rises. Moreover, it will have significant effects on long-term planning. In this regard, short-term power system modeling considers the operational and dispatch characteristics of the generation units. That considers the technical constraints of the conventional and renewable generation plants and the financial constraints such as gas price, cost of shut-down and start-up, and variable operating & maintenance cost (VOM). The output of short-term planning could vary based on the study itself. It can include system stability under a high share of renewable energy, voltage drop, rate of curtailment of renewable energy plants, and congestion in the power system network. The short-term variability of supply and demand is not the core focus of long-term energy planning models because they usually do not provide a high temporal resolution. This is why a separate suite of electricity market, unit commitment (UC), and dispatch models (UD) are specifically designed to investigate these issues [15]. Sometimes, these models are referred to as power system models. Tools in this category, like GTMax, PLEXOS, and Grid View, entail a much higher temporal resolution and cover numerous operational constraints. However, they might only provide limited insights into the design of future energy systems. A study by McIlwaine et al. [16] used PLEXOS to investigate the effect of battery storage deployment on a daily system operator's economic dispatch. They considered four combinations of wind and load in Ireland's isolated power system. The deployment of battery energy storage can reduce the total cost of generation, emission levels, and ramping rates. Shu et al. [17] explored the benefits of flexible electric vehicle (EV) charging in the year 2037 on Indian power grid infrastructures using unit commitment and hourly and sub-hourly dispatch models. They concluded that inflexible EV charging would lead to a substantial need for additional generation capacity. Dalala et al. [18] investigated the short-term effect of the high share of renewables on the Jordanian national grid's dispatch operation and running cost. Their PLEXOS results showed that high penetration of renewables changed the optimum network operation and increased the generation cost. Thus, increased energy storage is highly recommended to meet the load demand economically. Li et al. [19] discussed the optimum location and size of energy storage system under high share of renewable energy. Outputs result in improving the voltage stability and reduce the uncertainties associated with high renewable penetration.

Improving the depiction of solar and wind short-term changes in long-term energy models has gained popularity in recent years. Various techniques for incorporating short-term variations into long-term planning have been proposed. Soft linking [20] is considered the main adopted methodology for linking short-term operational analysis in long-term energy planning models. The results of the long-term analysis (i.e., Annual installed capacities of each generator type) are used as inputs for the short-term models to verify the operational constraints are met on an hourly basis for each generator type. Another rarely used

method is embedding the short-term operational constraints in the long-term model to work simultaneously toward convergence [21]. This method ensures that short-term constraints are met on an hourly basis while planning the long-term strategy. The difference in time scales between short- and long-term models makes the combined analysis complicated and time-consuming. This is why soft linking is widely used, and consequently, it will be adopted in this work.

Soft linking between long-term energy models and short-term models is widely employed in previous studies [22–25]. Welsch et al. [26] examined three distinct power sector models in Ireland, namely a long-term energy model called OSeMOSYS with twelve time-slices, an enhanced OSeMOSYS model incorporating technical constraints, and a soft-linked TIMES-PLEXOS model with 8784 time-slices. Their investigation revealed that the simplistic OSeMOSYS model tended to overestimate the requirement for system flexibility while exaggerating wind energy utilization. However, by introducing operational constraints, the modified OSeMOSYS model successfully replicated the outcomes of the soft-linked TIMES-PLEXOS model. Poncelet et al. [27] assessed the performance of a low-resolution TIMES (twelve-time slices) model applied to the Belgian power system. They used an hourly optimal dispatch model to examine the results achieved with and without specified technical limitations. They proved that for low VRE penetration (less than 25 %), the TIMES model's limited technical elements were a significant source of inaccuracy. As the VRE penetration threshold increased, the error from low temporal resolution became more prominent. Notably, at a 50 % VRE penetration level in the power mix, the dispatch error exceeded 10 %. This error manifested as an overestimation of VREs and base-load power plant production, coupled with an underestimation of dispatchable power plant output. Consequently, operational costs were underestimated by approximately 50 %. Deane et al. [20] established a correlation between TIMES and PLEXOS models, specifically applied to the Irish power system in 2020. This correlation was subsequently utilized as a proof of concept [28] extending its application to assess the security and evolution of the Italian power system until 2030. Through optimal dispatch analysis using PLEXOS, studies presented numerous intriguing findings regarding the optimal generating portfolio obtained using TIMES. However, they cautioned that this portfolio lacked sufficient reliability when considering the potential for load loss. Another noteworthy observation was the limited operational activity of many gas plants throughout the year, which raised concerns about their viability. Golombek et al. [29] soft-linked LIBEMOD, a multimarket energy equilibrium model of Europe, and TIMES Europe to provide an analysis of the decarbonization of the electricity sector up to 2050. To accommodate the short-term uncertainty of intermittent wind and PV supply, the transmission grid expansion is required to enjoy low PV costs fully. Zhang et al. [30] proposed a soft link between the long-term model and the operational model with a 1-h resolution. Under various nuclear evolution assumptions, they investigated the least-cost and least-CO₂ emissions options until 2030. One of their conclusions is that, depending on the target, there is a trade-off between coal and gas, and higher VRE integration (i.e., less curtailment) with more gas and less coal since gas plants are more dispatchable.

Egyptian power sector modeling has a few works in literature so far, but short-term energy planning has not been studied before for the Egyptian power sector. In 2015, Egypt launched its integrated energy strategy 2035 [31]. According to the 2035 strategy, renewable energy share in the capacity mix will reach 42 % in 2035. The capacity share of renewable energy (RE) has been extended to 50.4 % by 2040 [32]. This update is still under approval. Rady et al. [33] used OseMOSYS to evaluate the economic and environmental indicators of the Egyptian power sector until 2040. They found that combined cycle, wind, and solar PV are the most compatible technologies for future expansion in the Egyptian power sector. Research by Ref. [34] investigated the possibility of increasing the RE share in the Egyptian power system using nationwide canal-top PV systems. The proposed PV system over

irrigation canals was optimized to cover increased demand while reducing water evaporation rates. Mondal et al. [35] presented a TIMES model for long-term energy planning in Egypt's power sector. The model uses different policy scenarios to predict the installed capacity and the generation mix until 2050. Results show that implementing different scenarios could significantly change the capacity and energy mix. Dallmann et al. [36] investigated the potential pathway of the Egyptian power sector considering transmission network expansion to accommodate the future power system capacity. They used OSeMOSYS for deploying the model. The study concluded that renewable energy resources could shoulder a large share of the demand profile. However, it will require huge grid investment. Table 1 summarizes some recent research on the Egyptian power system, along with included features.

As outlined in Table 1, there is a clear gap in the previous studies concerning the short-term analysis specific to the Egyptian power system. To the author's knowledge, there is no short-term analysis that considers unit commitment operational constraints and relates that to the long-term energy planning that has been carried out. So, there is no verification for the long-term energy plan on an operational level under high penetration of renewable energy. Accordingly, the calculation of the optimal generation mix lacks operational accuracy. In addition, asset management is absent in most published studies. Generation companies need to build their future strategy to ensure not being subjected to substantial financial losses. The current work is original research focused on filling the literature gap with a clear methodology. Here, the short-term operational analysis of the Egyptian power system is performed on the generation level and soft-linked with the long-term modeling developed using the TIMES/Egypt model. This TIMES/Egypt model was established earlier during the development of the updated version of Egypt's energy strategy up to 2040 to calculate the optimum generation mix that achieves the least energy cost under high penetration of renewable energy sources. The study aims to verify that the performed least-cost long-term energy planning is consistent with operational constraints on an hourly basis using real operational data from the Egyptian system.

The contribution lies in developing a methodology for discrete and dynamic soft linking between the long-term planning and the short-term analysis, based on system stability and reliability criteria. Then, a detailed short-term analysis is performed based on the long-term planning outcomes given the power fleet's operational details. A set of economic, technical, and environmental indicators is proposed and calculated to judge the performance of the power fleet on an hourly basis (on the generation level) and represent measurable feedback. This feedback is then used to judge the adequacy of the long-term planning. In more detail, the PLEXOS power system modeling software tool is fed with outputs of the long-term planning (TIMES/Egypt) for 2020:2040. These inputs include potential RE capacity, the renewable resources' allocations, and the existing power fleet composition and operational

Table 1
List of recent work on the Egyptian power system.

Ref.	Long-term power system planning	Short-term Power system planning	Transmission network modeling	Year	Software model
[36]	●	O	●	2022	OseMOSYS
[37]	●	O	O	2022	Agent-Based Modelling
[38]	●	O	O	2020	New generation expansion planning model using MATLAB
[35]	●	O	O	2019	TIMES
[33]	●	O	O	2018	OseMOSYS
[39]	O	O	●	2017	Adaptive Neuro-Fuzzy Inference System

details of each generator. The hourly performance of each generator in the power system is verified to satisfy the operational constraints for one year. In addition, the generation mix must satisfy the minimum short-run marginal cost (SRMC), with no or minimum curtailment. Five sample years are selected for the analysis, including the reference year 2020. The operational short-term analysis considers unit commitment operational constraints such as; conventional plants' ramp rate, start time, shutdown time, hot, intermediate, and cold start times, minimum uptime and downtime, and minimum stable level. Several conventional and renewable generation technologies are included in the analysis, in addition to pumped hydro storage (PHS). Furthermore, hourly output for both wind and solar plants has been considered using WindPro [40] and SAM [41] software tools, respectively. This study aims to verify the long-term energy plan on an operational level under high penetration of renewable energy. Also, it aims to identify the necessary energy storage to balance the grid and find the optimal generation mix that achieves the least short-run marginal cost (SRMC). Finally, this analysis can be used as a base for an asset management program for the power sector in Egypt. The novelty of the research lies in being the first detailed study on the operational level of the Egyptian power system that uses discrete soft-linking and various performance indicators. The Egyptian power system is one of the largest systems in the region, with substantial input data, which can be used as a guide in many other systems. The study relies on the most updated real data from the Egyptian grid operator. Also, the hourly renewable energy generation is calculated based on the Egyptian strategy and the updated combined renewable energy master plan of Egypt, which is the official document for renewables potential in Egypt. The linking with long-term energy modeling opens the way for more sectoral integration and comprehensive studies. In addition, the analysis is performed using widely available software tools that generalize the methodology beyond the studied case's geographic constraints.

2. Methodology

This section describes the methodology and modeling approach for short-term planning and soft-linking with long-term planning. As mentioned earlier, the target is to verify the long-term planning using

the power system's short-term unit commitment and economic dispatch characteristics. Fig. 1 shows the methodology flowchart followed in the current study. The process starts with getting the long-term energy plan from the TIMES/Egypt model created during the preparation of the Egyptian energy strategy. Then, allocate the renewable resources needed to cover the renewable energy capacity addition in the long-term plan up to 2040 (shown on the side map). Given the allocations of renewable resources and the operational characteristics of the generation fleet, PLEXOS power system modeling software will perform the short-term analysis of the Egyptian power system. PLEXOS will check the unit commitment and economic dispatch constraints and calculate several indicators, such as SRMC and curtailment of renewable energy. If the short-term constraints are satisfied, under the current long-term scenario, then the long-term planning is adequate, hence, the generators' performance is optimized, and the indicators are calculated. If the short-term constraints are violated in one or more generators, an effort is exerted to accommodate the long-term planning by redistribution of roles between different generation units. If maneuvering didn't work in any way, feedback is returned to adjust the long-term planning (i.e., capping the renewable share), and the process is repeated till short-term constraints are satisfied, as indicated in Fig. 1. Consistency between both the TIMES and PLEXOS results is ensured by using the same annual average values for the inputs. Furthermore, the total output values of both the long and short terms are cross-checked, where in the short term RE is used to supply the load, and the curtailed energy and losses in the storage systems are balanced with that predicted in the long term. On the other hand, the lost load value is set to 10000 USD/MWh. Hence, this system always tries to look ahead and optimize the operation of the conventional fleet to avoid the lost load price. In addition, a reserve capacity (equal to the largest conventional unit + 50 % of the renewable fleet) is included for any possible uncertainty in the system. As a test of the validity of the methodology, a long-term scenario with 53.2 % renewable capacity penetration (with 49.2 GW PV capacity) was modeled on a short-term basis. The power system, as per PLEXOS results, was not stable. It was found that the spinning reserve was not achieved, and there was excessive cycling of the conventional fleet. Accordingly, the long-term plan was adjusted to solve these problems,

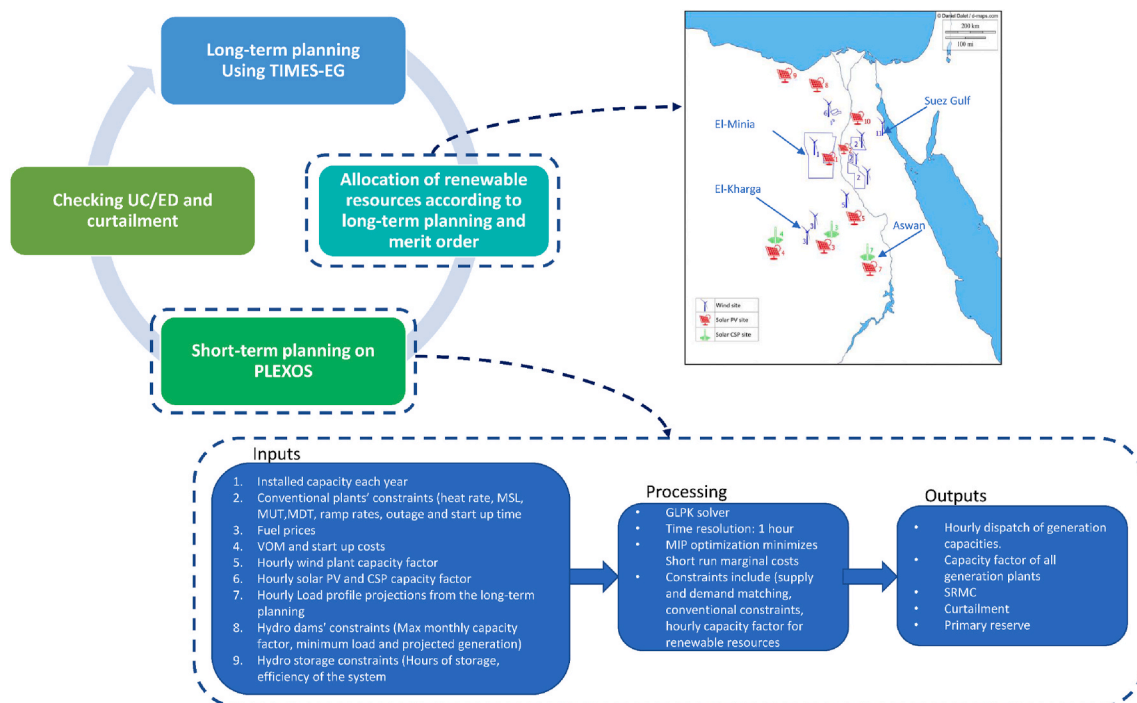


Fig. 1. Methodology Flowchart (Modelling framework).

leading to the current long-term scenario. The first long-term scenario, that didn't achieve the operational constraints, is presented in Appendix A for reference. The second updated scenario (i.e. the current approved long-term plan) is the scope of this study. Section 2.1 introduces the PLEXOS Integrated Energy Model and Section 2.2 describes the long-term planning outcomes which are fed to the 2020 reference year and the test models in 2026, 2030, 2035, and 2040.

2.1. PLEXOS model

PLEXOS software is a power system modeling tool used for power system modeling including both long-term and short-term planning [42]. Also, PLEXOS is a market simulation software. It can be used for almost all types of utilities including electricity, gas, and water. PLEXOS is used to solve unit commitment and economic dispatch UC/ED models. unit commitment (UC) is a class of mathematical optimization problems in which the production of a group of generators is coordinated to achieve certain goals, such as minimum operational cost, where economic dispatch (ED) is defined as the operation of the highest efficiency unit to achieve a low operational cost [43]. The software has several solvers to solve mixed integer problems. In this study, GLPK solver is used to solve the economic dispatch and unit commitment problems. GLPK is a free package intended for solving mixed integer programming (MIP) and other related problems [44]. The UC/ED problem solved by PLEXOS has an objective function as shown in equation (1). Many technical constraints are considered for the validity of the model such as: minimum stable level for the generation units, minimum ramp-up time, minimum downtime, ramp rate, and hot/intermediate/cold start-up time. Table 2 lists the technical constraints used in the analysis [45]. It is important to mention that, unlike Times which considers only aggregated technologies, PLEXOS models each unit alone. This means that a plant that consists of four units could operate only two units and shut down the other two. Conventional plants' heat rate was obtained from the annual report of the Egyptian Electricity Holding Company, EEHC [46].

$$\min \left\{ \sum_{t=1}^T \left\{ \sum_{i=1}^N v_i c_i p_i(t) + v_i^s S_i \right\} \right\}, v_i \in \{0, 1\} \quad (1)$$

Where:

T is the time step, T is the total time interval, I is the generator, N is the total number of generators, v_i is a binary variable indicating if the generator is selected or no. c_i is the operational cost of the generator. $p_i(t)$ is the power generated by generator p_i at each time step. v_i^s is the binary variable indicating the startup of the generator. S_i is the startup cost of the generator.

In addition to the previous general technical constraints, some other special technical constraints are included in the analysis. Nuclear power plants are considered as baseload power plants, so minimum load is strictly specified.

Hydro dams are divided into two groups as shown in Fig. 2. The main hydro dam in Egypt is the High dam and it is modeled based on a prescribed seasonal flow variation taken from historical data [47]. Maximum capacity factor is prescribed for each season (summer and winter) and maximum energy production per year is applied so the

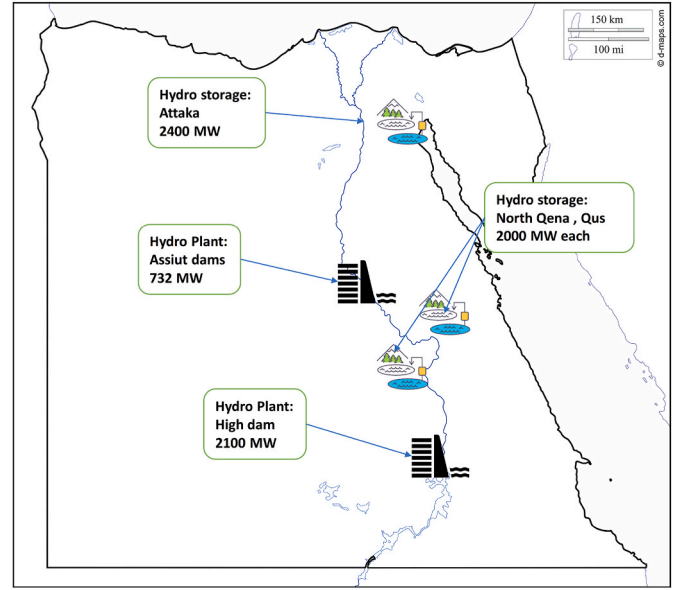


Fig. 2. Hydro and Hydro storage locations in Egypt up to 2040.

system cannot exceed it. The High dam is used as a peaking plant, as it enjoys a downstream basin between it and the following Hydro plant, which is Aswan Dam. This enables to operate it as a peaking or balancing plant of capacity 2100 MW. Other dams such as the Assiut dam are not allowed to store water and pump it when needed. So, all dams on the Nile River except the High dam are modeled to be working all day with their allowable capacity.

Pumped hydro storage systems are modeled as two reservoirs (upper and lower) and a cycle operation between them allows the pumping and generation modes of load. The capacity of pumping and turbine is considered with a conversion efficiency of 70 % and 10 h of storage. Table 3 summarize the technical constraints for these types of special generation. To reduce the modelling complexity; the transmission network is considered a copper plate which means that congestion and curtailment due to the transmission network are not considered in the analysis. This is a common technique to reduce the computational burden with minimum impact on generators' performance [48].

The short-run marginal cost (SRMC) for each generator is calculated based on equation (1). According to the economic dispatch criterion, the model first operates the lowest SRMC units, then higher ones until fulfilling the required load.

$$SRMC = VOM + Fuel\ cost + Start\ up\ cost \quad (2)$$

VOM costs are the portion of the O&M costs that are a function variable of the level of energy production (MWh). Table 4 lists the VOM [49] used in the analysis, considering the reference year and taking a yearly inflation of 8 %.

Current conventional generators are assumed to be fully operated by

Table 2
Technical constraints for conventional generators [45].

Technology	Minimum stable level (%power)	Ramp rate (%power/min)	Minimum up time (hr)	Minimum down time (hr)
CCGT	50 %	10 %	6	6
OCGT	50 %	20 %	1	1
SPP-G	40 %	1 %	6	6

CCGT: Combined-cycle gas turbine, OCGT: Open-cycle gas turbine, SPP-G: Gas-fired steam power plant.

Table 3
Special technical constraints for nuclear and hydro plants.

Technology	Constraint
Nuclear Power Plant, NPP	Must-Run Unit
Hydro	Minimum load factor: 90 % High dam: Seasonal flow: 30 % monthly capacity factor for winter months and 65 % monthly capacity factor for summer months. Minimum load: 150 MW for winter months and 500 MW for summer
Hydro storage	Other dams: 723 MW, Working as baseload. Overall efficiency: 70 % Storage time: 10 h

Table 4
Projected VOM [\$/MWh] for different generators up to 2040 [49].

Variable O&M	2020	2025	2030	2035	2040
CCGT	0.38	0.6	0.82	1.2	1.77
OCGT	0.82	1.3	1.77	2.6	3.82
SPP-G	0.32	0.51	0.7	1.02	1.5
Pumped hydro	0	0	0	0	0
Hydro	0	0	0	0	0
CSP	0.26	0.41	0.56	0.82	1.2
PV	0	0	0	0	0
Wind	0	0	0	0	0
Nuclear	1.96	3.11	4.23	6.22	9.14

natural gas. Gas price is obtained according to 2020 prices [50]. The 2020 price is 3.08 \$/GJ. Applying a conservative subsidy lifting based on the cost of supply approach and based on the governmental plan, the projected incremental increase rate of 0.25 \$/GJ per year is considered. The price will reach 4.26 \$/GJ in 2025, and 5.2 \$/GJ in 2030, which is the expected cost of supply of the locally produced gas [50]. Then it is fixed throughout the rest of the analysis at 5.2 \$/GJ. Additional expenses are incurred when a thermal power plant is starting up. The equipment is usually heated up during the start-up, which can take several hours to produce the high-pressure steam needed to power the steam turbine. Several startup costs are defined in the analysis based on the data enabled by EEHC including hot, intermediate, and cold startup costs. Table 5 summarizes the startup costs used in the analysis [45].

In 2020 the total wind installed capacity is 1400 MW [51]. According to the New and Renewable Energy Authority, NREA [51], 1600 MW of solar PV, and 140 MW of solar CSP are in operation by 2020.

A recent study carried out by Ref. [52] developed the Egyptian renewable energy master plan. According to the study, they developed a renewable energy potential map for Egypt that covers Wind, PV, and CSP. The results of this study include a techno-economic feasibility study for each site and the hourly production of renewable energy from each site. Using these data enabled us to model renewable energy correctly in the short-term analysis. Table 6 lists the potential generation of renewable energy in selected areas in the Gulf of Suez, Minia, El-kharga, and Aswan. Fig. 3 shows the average diurnal wind production at each site. A site like Minia has a huge capacity credit value as it has more energy generation during peak periods and less generation during midday when solar power is peak. On the contrary, Suez Gulf and El-Kharga have high output during the day, and low energy output during night.

For solar PV, Fig. 4 shows the average diurnal energy output from each site. It's an almost identical pattern for all sites. The curves have a sharp ramping up and ramping down due to the single-axis tracking system used to increase energy production. The only drawback from a power system point of view is the sharp ramping up and down. This drawback is shown later in the results when conventional generators have to ramp up fast to compensate for the decrease in solar PV during sunset hours.

For CSP, Fig. 5 shows that both Aswan and El-Kharga plants have approximately equal annual output. The CSP plant is modeled for 8 h of storage, so it produces low energy during the day to store the rest of the energy for night operation, where demand is high.

The 2020 hourly demand profile is calibrated based on 2017 demand profile [53] for Egypt, given the total energy consumption and the peak load for 2020 [54]. This calibration assumes that the diurnal demand

Table 5
Cycling cost of conventional generators [45].

Unit type	Start-up cost [\$/MW]
Combined cycle power plant	45
Steam power plant	83
Gas power plant	40

Table 6
Average capacity factor for renewable energy plants.

Site	Wind capacity factor	Solar PV capacity factor	Solar CSP capacity factor
Suez Gulf	55 %	Not modeled	Not modeled
El-Minia	53.2 %	Not modeled	Not modeled
East Minia	45 %	29.2 %	Not modeled
El-Kharga	60.4 %	31.4 %	68.8 %
Aswan	Only for solar	30.9 %	68.9 %

profile is invariant for the period of study. Fig. 6 shows the diurnal load pattern of Egypt. As shown in the figure, the system has a peak load at night, which will limit the capacity of solar energy, as in this case, the capacity credit of solar energy is zero. It also impacts the ramping up and down of conventional generators.

PLEXOS solves a yearly UC/ED problem using a rolling horizon approach for one year with a schedule of one week and an hourly step. So, the solver will run 52 optimization processes with a resolution of 1 h with 6 h looking ahead for prediction of solar and wind patterns, and dispatch conventional generators according to that. Also, forced outages are included in the analysis. Outages can occur at any time of the year, according to the Monte Carlo simulation. The model also includes scheduled maintenance. Based on the yearly maintenance rate and mean time to repair, the model predicts the ideal time to conduct regular maintenance [47].

2.2. Long-term model outcomes

Based on work done on TIMES-EG, a long-term plan is proposed for Egypt 2040 [32]. However, the long-term plan does not consider all the operational constraints that could happen on an hourly basis. Hence, a short-term planning is needed to validate the long-term planning. According to the long-term plan (which considered the expected growth rate in the national Gross Domestic Product (GDP), sectorial GDP, energy prices, growth in population, technology development, and energy efficiency measures), Egypt's peak load is expected to reach 66.9 GW [32]. Fig. 7 shows the evolution of the peak load from 2020 to 2040.

The year 2020 is the first year in the long-term model results, Hence, it's used as a base year to verify the model results with the actual data from EEHC. According to the annual report by EEHC for the year 2019/2020 [55], the total installed capacity in 2020 is 59.4 GW [46]. The generation pool consists of 31.9 GW of combined cycle plants, 18.1 GW of Steam plants, 3.5 GW of gas turbines, 2.8 GW of Hydro plants including 2.1 GW from the high dam and 0.7 GW distributed along the Nile as shown in Fig. 2. Fig. 8 illustrates that the renewable generation portfolio includes 3.14 GW of wind, solar, and CSP farms and plants.

After validating the 2020 base scenario, four different milestone years are considered in the short-term model to validate the long-term plan. The years under consideration are 2025, 2030, 2035 and 2040. Fig. 9 shows the renewable penetration (not including hydro plants) as a percentage of capacity according to the long-term planning outputs. As shown in the figure, renewable penetration slightly exceeds 50 % of the total installed capacity in 2040.

Fig. 10 shows the projected development in the installed capacity up to 2040, according to the long-term plan. As shown in the figure, the total installed capacity of CCGP will increase by only 2.8 GW from 2020 to 2040; this is the balance between decommissioning and commissioning plans. Steam power plants' installed capacity will decrease by 6.9 GW. The nuclear power plant at Al-Dabaa north Egypt is already under execution with a capacity of 4800 MW [56]. This nuclear capacity is planned to enter the service by the end of 2028 for the first unit and 2030 for the last and fourth units. Hence, nuclear power plants (NPP) started to be included in the simulation by 2030. Renewable's capacity penetration including hydro plants has increased from 4.8 % in 2020 to 50.5 % in 2040.

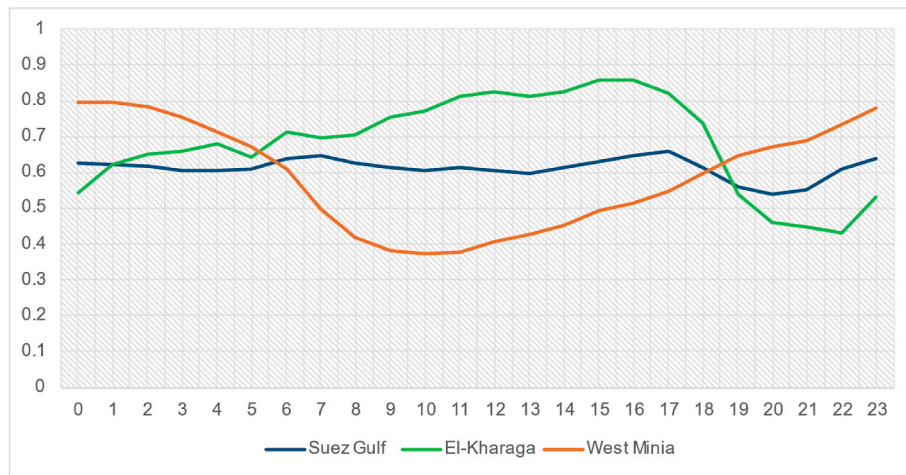


Fig. 3. Normalized annual diurnal energy production for wind energy potential [52].

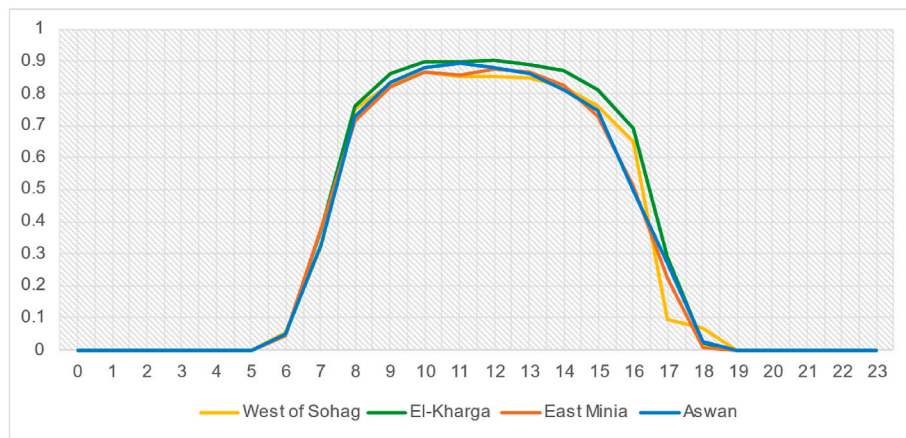


Fig. 4. Normalized annual diurnal energy production for PV energy potential [52].

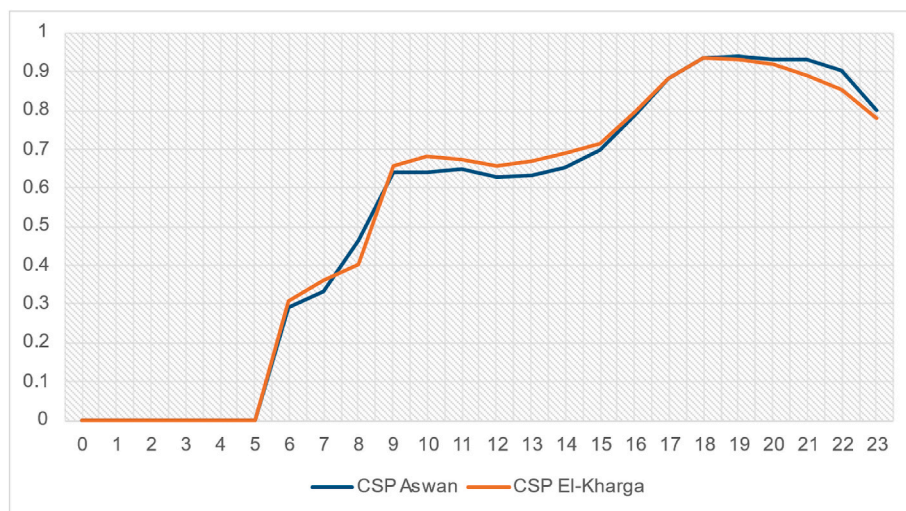


Fig. 5. Normalized annual diurnal energy production for CSP energy potential [52].

Due to the high penetration of renewables in the power system, measures are taken to ensure grid stability with such high renewables share. A 6.4 GW of pumped hydro storage is planned for energy storage. As planned, the first 2400 MW pumping and storage hydro project is

planned to be constructed in 2026 [57], while the remaining two plants each with 2000 MW are planned to be commissioned before 2035. Also, the Egyptian-Saudi interconnection is planned for commissioning in 2025 [58], thus it will add a total interconnection capacity of 3000 MW,

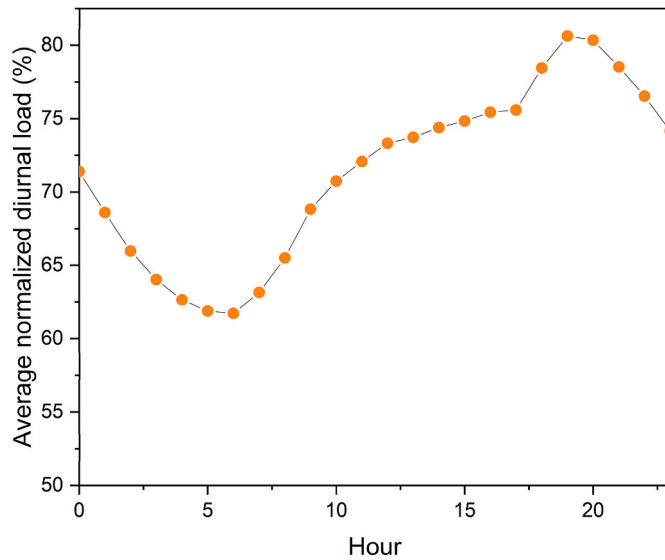


Fig. 6. Normalized diurnal load profile for Egypt.

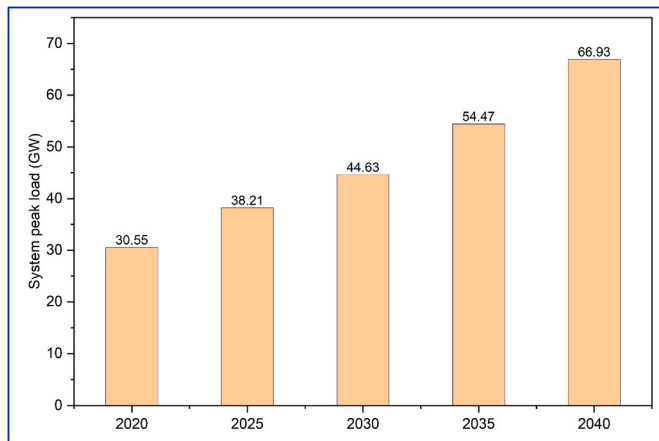


Fig. 7. System peak load [32].

yet the average exchange capacity has been considered in the analysis to be 1.5 GW and losses of 2 %. This interconnection is supposed to support the Egyptian grid at night when the peak demand occurs and support the Saudi network during the midday during the Saudi system peak time.

3. Results and discussion

In this section, the five milestone years under investigation are analyzed to find out, if the long-term plan up to 2040 can be achieved on

the short-term level (hourly operation level), such that the system dispatch limits are not breached and the system stability and reliability are ensured as well as no or minimum curtailment for the renewable energy plants is needed. As explained, this analysis is carried out using PLEXOS software, where the mixed integer programming optimization is used to minimize the operating costs. Also, unit commitment (UC) and economical dispatch (ED) parameters are considered with a resolution of 1 h. In this analysis, the transmission network is considered a copper plate, which means that no congestion/curtailment due to transmission constraints is taken into consideration. The impact of operational constraints such as Minimum stable level, ramping rates, hot/warm/cold startup times, and cost and minimum/maximum up time are considered.

3.1. Energy mix and operational scenario

According to the short-term planning up to 2040, using PLEXOS software, Table 7 illustrates the total generation from each technology. As presented in Table 7, nuclear power plants start to operate in 2030. Also, increasing penetration of renewables limits the total generation from the thermal units. Hydro storage and interconnections are essential for the balancing of the system under a high share of renewables. That's because of the high response rate of these units. Hydro storage plants' capacity factor is 13.5 % on average in 2040. While it's 4.4 % in 2035. This is due to the increasing penetration of renewable energy in 2040 and the need to store the excess renewable energy and discharge it when it's needed. Fig. 11 shows the installed capacity and energy generation from the pumping and storage facilities through the milestone years. The Egypt-Saudi interconnection capacity factor is 28 % in 2040. While in 2035, it's only 19.5 %. Also, the share of gas turbines in the energy mix has increased in 2040, due to the high penetration of renewables and the need to compensate for the uncertainties in renewables generation.

Renewable generation has increased from 10.6 TWh in 2020 to

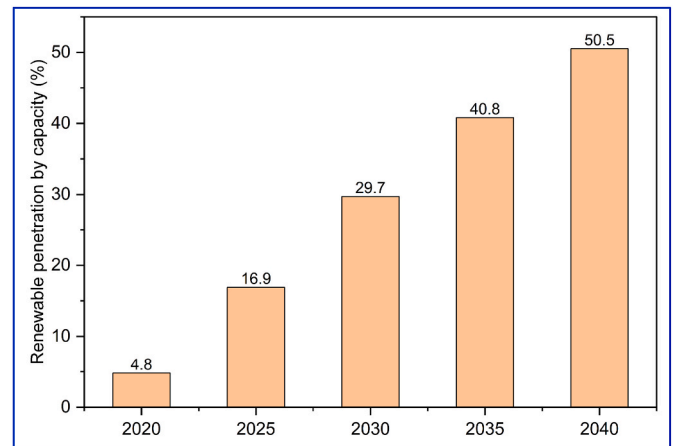


Fig. 9. Renewable penetration by capacity percentage [TIMES-Egypt] [32].

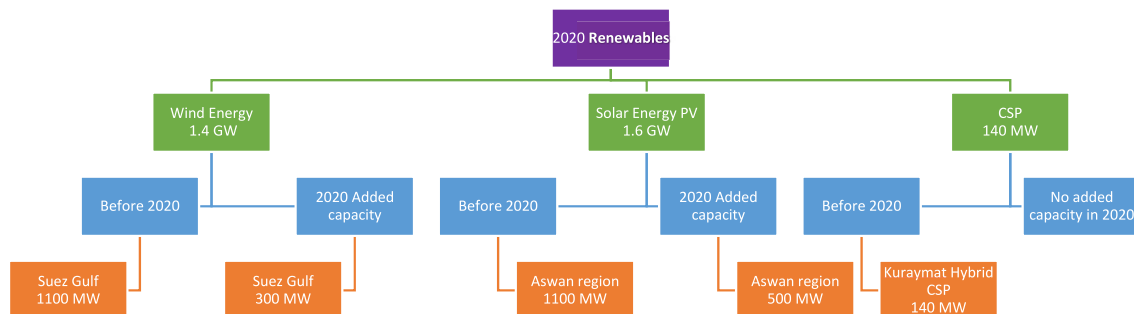


Fig. 8. Renewable capacities in the 2020 base year (not including hydro) [51].

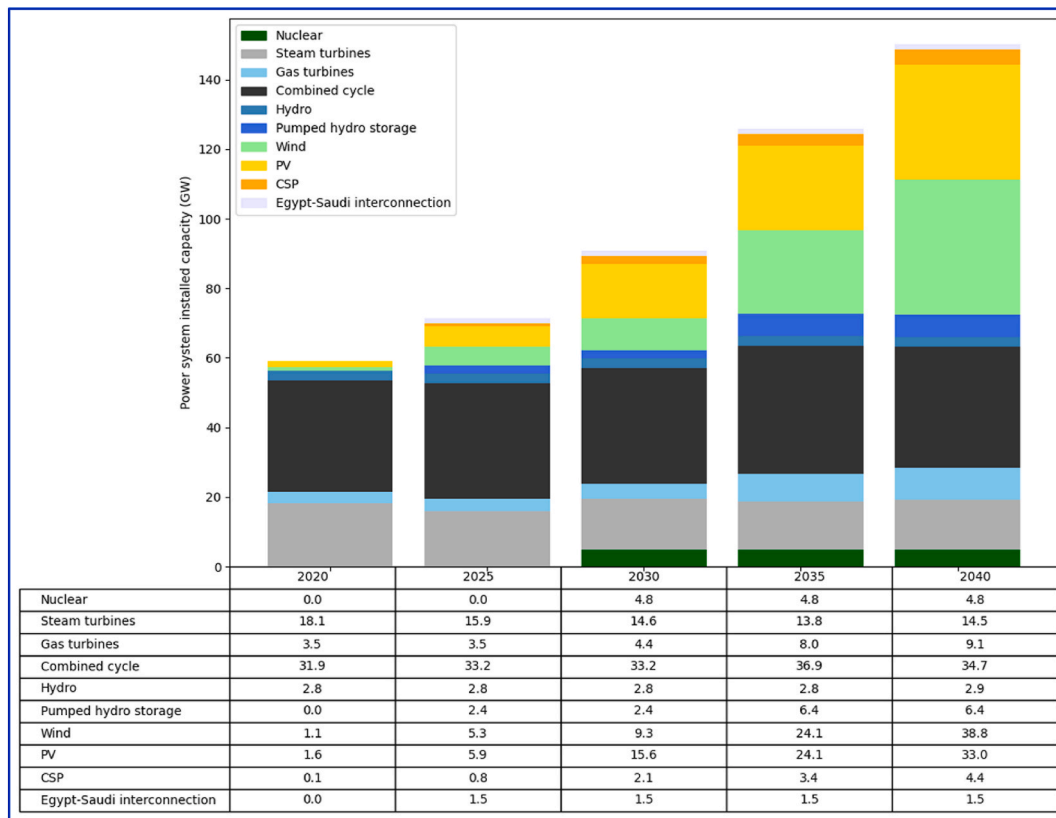


Fig. 10. Long-term planning results: capacity mix [32].

Table 7
Electricity mix for each year on PLEXOS.

Technology	Electricity Mix (TWh)				
	2020	2025	2030	2035	2040
Nuclear	–	–	41.93	41.91	40.67
Combined Cycle	210.73	183.58	132.22	113.11	71.79
Steam Turbines	3.22	–	–	–	0.04
Gas Turbines	0.12	0.05	0.25	0.35	1.12
Renewables	10.68	37.68	93.96	179.73	287.17
Hydro	13.42	12.52	12.52	12.54	12.53
Hydro Storage	–	–	0.50	2.48	7.58
Egypt-Saudi Interconnection	–	0.93	2.43	3.24	4.00

287.17 TWh in 2040. More than 28 times increase in the renewable generation. Fig. 12 shows the breakdown of the renewable generation. Another important aspect of renewable energy is the complementary nature between wind and solar energies. Most of the Egyptian potential sites for wind energy have a peak generation during nighttime, while solar energy has a peak during noon time. This complementarity between resources reduces the curtailment and energy needed for balancing [52].

Fig. 13 shows that the renewable energy share has increased from 4.5 % in 2020 to 69.5 % in 2040. As a result, many thermal plants will be phased out. Thermal plants' energy share will be reduced from 89 % in 2020 to 17.65 % in 2040. The share of steam and gas turbines is less than 1 % in 2040. As its only limited to the peak hours.

The generation mix reported from the long-term planning differs from the generation mix in the short-term planning. This could be attributed to many criteria. Renewable allocation is one of the criteria, as long-term planning deals with any renewable technology with the average value of the capacity factor. Hence, in some years, there might be higher capacity factor sites rather than low-capacity factor sites. Moreover, curtailment is not considered in long-term planning. In short-

term planning, curtailment of renewable energy is a key factor for determining energy mix and system stability. Figs. 14 and 15 show the difference between long and short-term in the analyzed system.

Figs. 16 and 17 show the hourly dispatched electricity from the power system for the lowest and highest demand days during 2040. Each figure shows the system's performance in two consecutive days in 2040. Fig. 16 shows that on the first day (January 4, 20240), there are high renewable resources available, so conventional plants are shut down during the daylight. The load is fully powered by nuclear and renewable resources. Starting from 3 p.m. (thanks to the Look Ahead function in PLEXOS), combined cycle power plants started up again to compensate for the coming deficiency in solar resources at sunset. However, as ramping up for a combined cycle is not enough to compensate for the lack of solar energy at 5 p.m., gas turbines, hydro, and hydro storage plants are ramped up as fast responders to cover the load for that hour. For the peak demand days (July 16–17, 2040), the system has a lower share of wind energy, hence, combined cycle power plants are operated during these two days, to compensate for this reduction in renewable generation.

Fig. 18 shows the energy output from hydro storage plants. The use of hydro storage is related to the peak demand hours or hours with low renewable production. In that case, hydro storage plants have peak production during peak hours when it can assist until the full ramping up of conventional generators. The same pattern can be depicted in Fig. 19, where hydro dams perform like hydro storage plants.

Fig. 20 shows the capacity factor of thermal units has been reduced due to the high penetration of renewables. The average capacity factor for combined cycle power plants has been reduced from 65.9 % in 2020 to 17.98 % in 2040. Also, it's obvious that the total share of steam turbine power plants and gas turbine power plants is very low due to the low efficiency and increased short-run marginal cost of these units. Because the economical dispatch of the system prefers to use combined cycle power plants over steam and gas turbines as it has higher

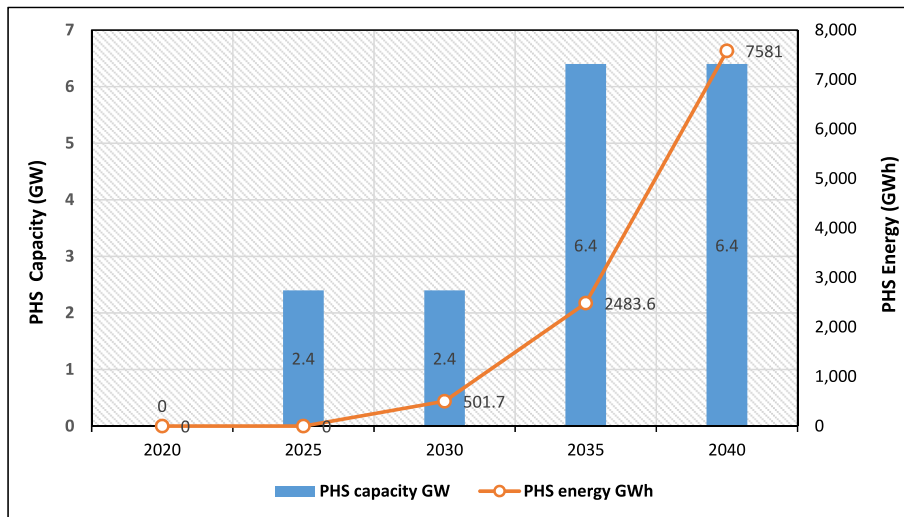


Fig. 11. Pumped hydro storage capacity and energy share.

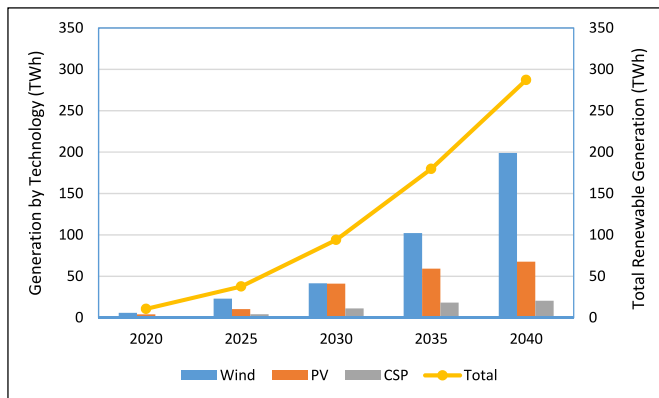


Fig. 12. Renewable energy generation per year.

efficiency.

3.2. Curtailment

Integrating intermittent renewable resources into power grids could be a challenge due to the mismatch between the renewable electricity supply and demand. This leads to curtailment of renewable energy sources at times of surplus supply. Although the installed capacity of intermittent renewables will reach 76.2 GW in 2040 [59], while the peak load will reach 66.9 GW in 2040, however, this will not be the reason for curtailment. The probability for curtailment in this case happens if all the renewable plants are available and operate at their peak output and coincide with each other, which is not the case. Curtailment will happen due to the impact of variations in the generation energy of the renewable plants and their impact on breaching the dispatch limits of the conventional plants. Or the increase of renewable generation such that it exceeds the allowable generation from other resources as considered by the must-run plants. Curtailment analysis

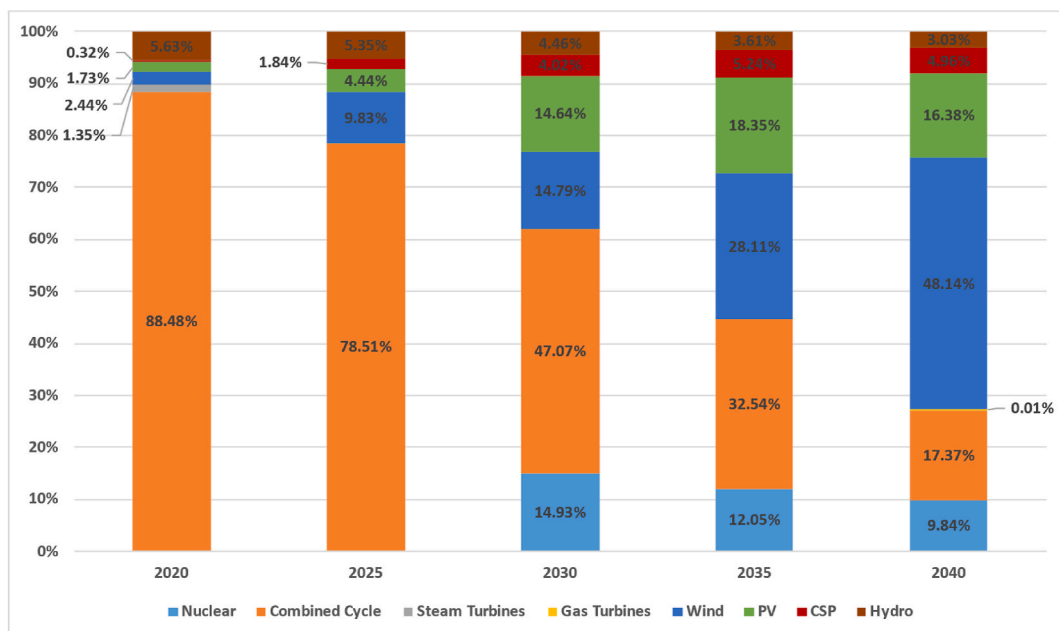


Fig. 13. The energy mix for each year.

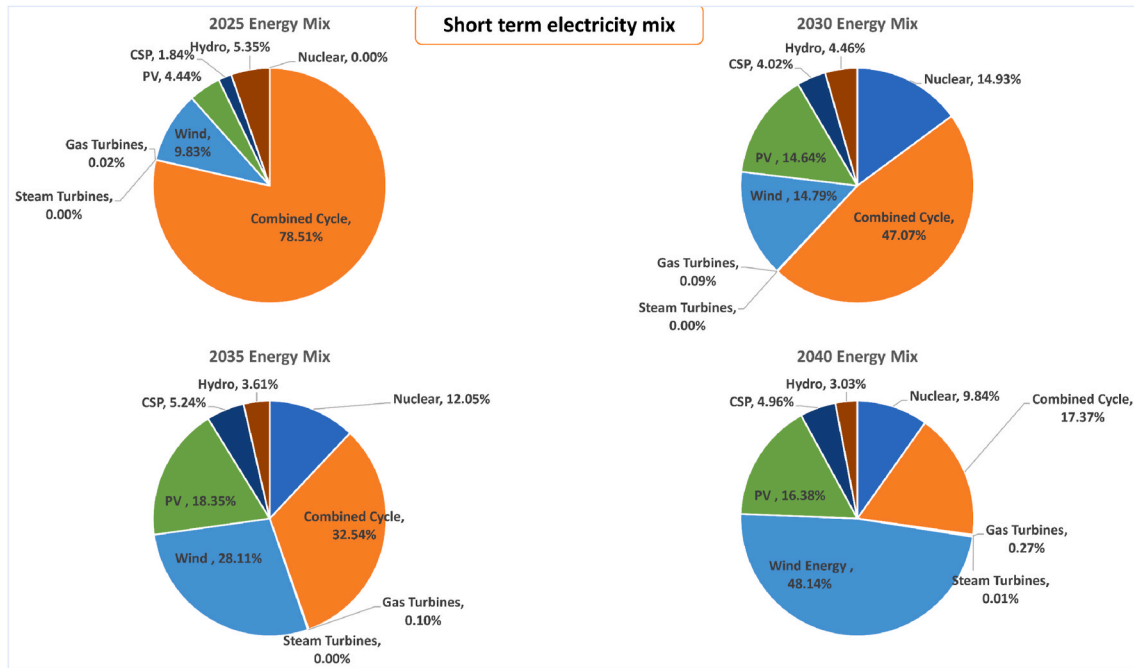


Fig. 14. Short-term electricity mix.

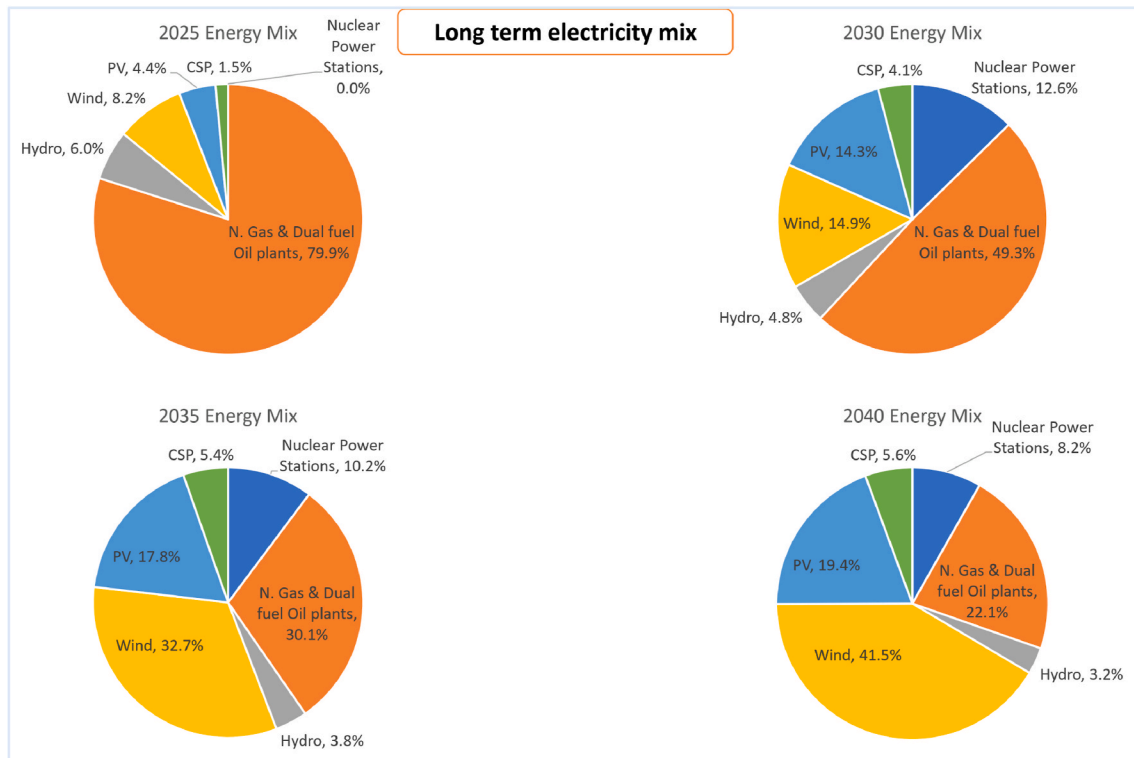


Fig. 15. Long-term electricity mix.

showed that the system will have no curtailment until 2035. However, for 2040 curtailment will appear in CSP, wind, and PV plants. Fig. 21 shows curtailment as a percentage of total energy share for each renewable plant technology at different sites in 2040. The system curtails first from the CSP plants, as it has the highest associated costs with the produced energy due to its high CAPEX. Then, curtailment occurs in both PV and wind, based on the availability of the resources themselves (i.e., site characteristics).

3.3. Average weighted short-run marginal cost (SRMC)

SRMC is defined as the incurred cost required to produce 1 MWh of electricity. The relationship between Average weighted SRMC and renewable penetration is more than complex. Increasing renewable penetration could cause the average weighted SRMC to increase as the required balancing energy to stabilize the grid could increase. Alternatively, increasing renewable penetration reduces conventional fuel

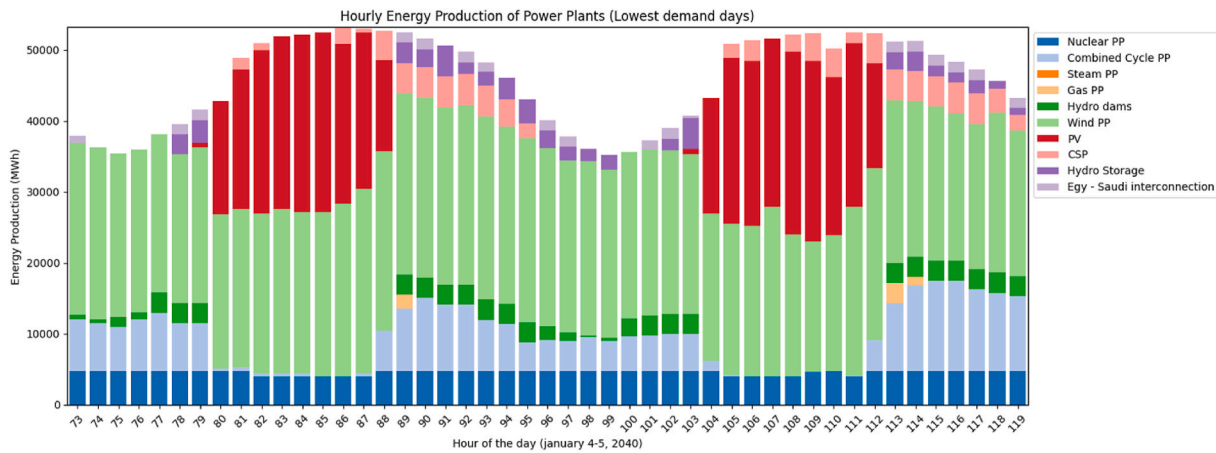


Fig. 16. Hourly dispatch for two days in January during 2040.

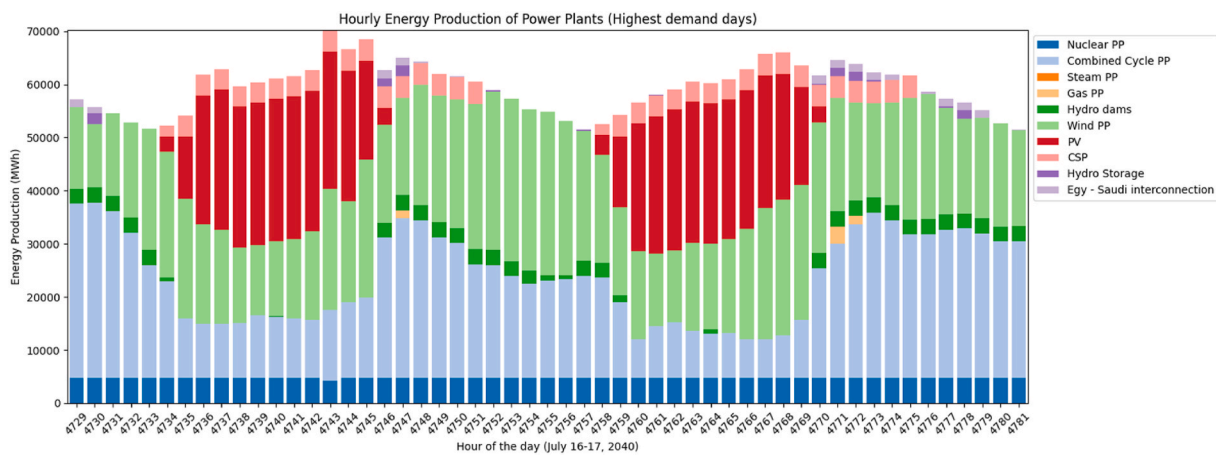


Fig. 17. Hourly dispatch for two days in July 2040.

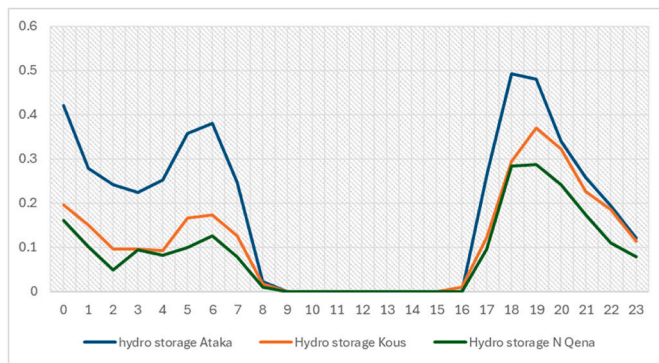


Fig. 18. Hydro storage plants normalized production.

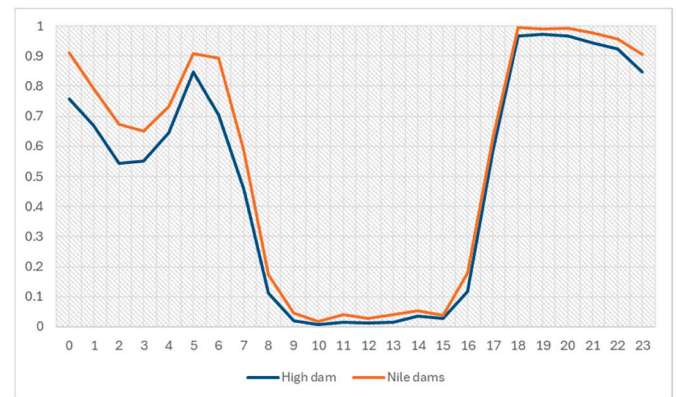


Fig. 19. Hydro plants normalized production.

consumption, which represents a major part of the running costs of the fleet. PLEXOS calculates SRMC for each plant. Average weighted SRMC is calculated according to the following equation (2):

$$\text{Average weighted SRMC} = \frac{\sum \text{SRMC for each plant}}{\text{Total energy produced}} \quad (3)$$

Fig. 22 shows the average weighted SRMC for each year in the analysis. As shown, the average weighted SRMC will be less down to 7.8 \$/MWh in 2040. This is attributed to the high penetration of renewables and the low operating hours of conventional units in 2040.

Fig. 23 shows the total system cost breakdown, fuel cost has the highest portion in all year. It reaches 86 % of the total system cost in 2040. VOM increases year by year due to the adjusted inflation of 8 %, as mentioned earlier. Fig. 24 shows that startup cost also increases due to the excessive cycling of conventional generators in 2040. The number of startups of conventional generators has increased exponentially from 2020 to 2040 to reach around 11,500 startups in 2040. Over 66 % of the total start-ups occur in the combined cycle power plants as the dominant technology in the conventional fleet due to their higher efficiency. Gas

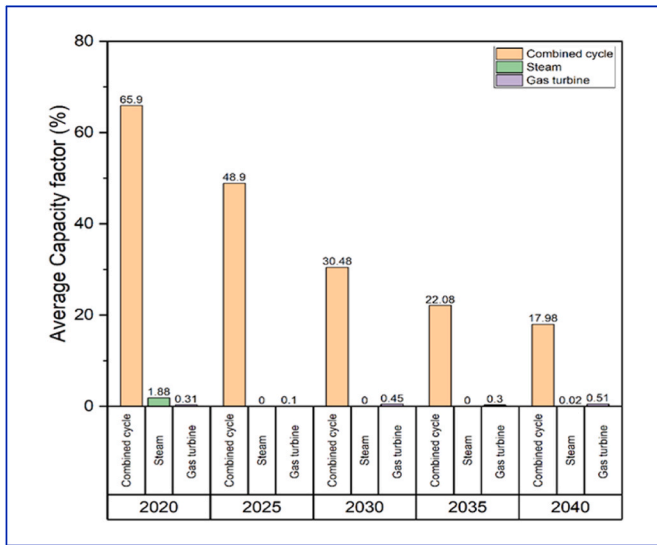


Fig. 20. The average capacity factor of conventional generators.

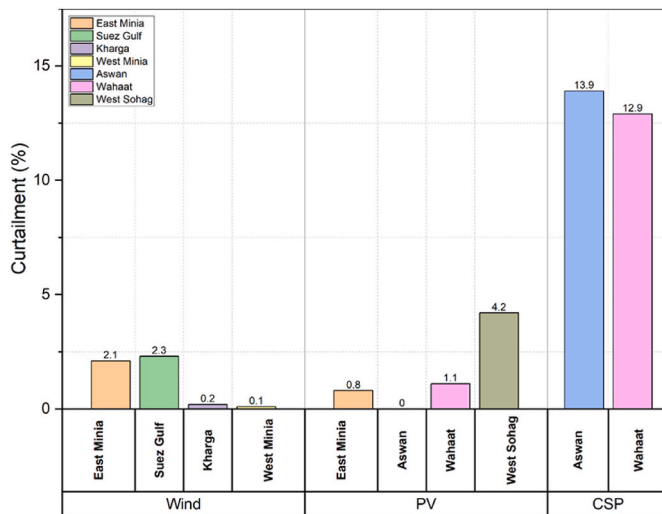


Fig. 21. Curtailment of renewable resources by technology at different sites during 2040 (percentage of total energy share).

turbines are more frequently used than steam turbines in 2035 and 2040, despite their lower sufficiency. This is due to their flexibility and lower startup time, which is needed in cases of high share of intermittent renewable sources.

3.4. Conventional fleet reliability

The conventional fleet of the Egyptian power system is crucial for grid stability. Fig. 25 shows the capacity share of the conventional fleet from 2020 to 2040. In 2040, the capacity share of the conventional fleet will drop from 90 % in 2020 to 39 % in 2040 due to the increased penetration of renewables.

Fig. 26 shows the required system ramp rate at the last hour of the day when the solar plants are phased out. Around 20 GW/h is required by 2040 at just 1 h to balance the system, this happens when the PV is going off at sunset. This is a huge number to be considered, especially if the network is modeled. The looking ahead function, available in PLEXOS, helps eliminate this issue by starting the conventional fleet 3 h before sunset. This ensures the smooth exit of PV units without disturbing the grid balance, yet this increases the operating cost.

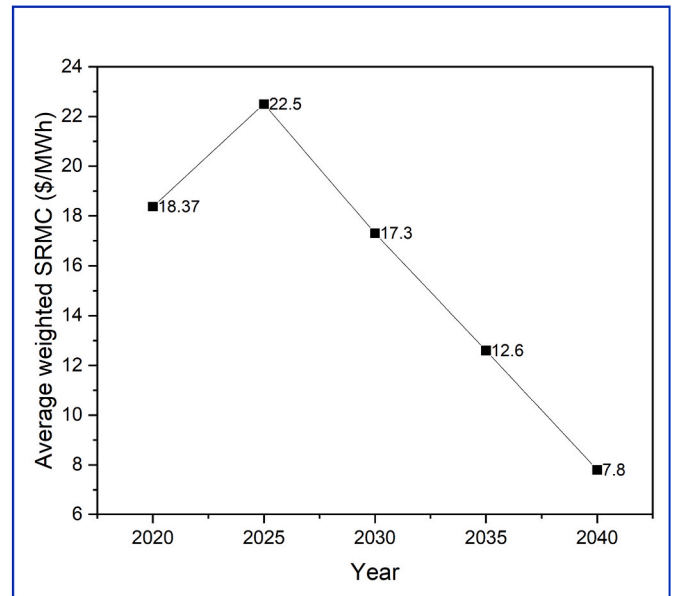


Fig. 22. Average SRMC calculated for the power system.

3.5. Reserve capacity

Primary reserve should take not only the possible shutdown of the largest conventional unit in the system but should also take into consideration the expected variation in the generation of the renewable energy units. The required primary capacity is equivalent to the largest conventional unit in the system plus 5 % of the total renewable plants' capacity. For 2040, this leads to a required primary reserve of 5000 MW, which is represented by 1200 MW of a combined cycle unit and 3800 MW of renewable capacity. Firstly, conventional generators are used to provide the required reserve of 5000 MW. However, due to the cycling of conventional generators, it couldn't be possible to maintain the reserve all the time. Hence, hydro storage facilities are used. Using hydro storage facilities to provide the reserve has been used widely in many previous studies, as presented in Ref. [60]. The lower part of Figs. 8 and 9 (presented earlier in sec 3.1) shows the primary reserve provision. The required 5000 MW mainly consists of two providers, the first is the combined cycle power plants (CCPP) and the second is the hydro storage systems. The provision of primary reserve from hydro storage units is larger than CCPP, as hydro storage provides reserve in both operations of charging and discharging.

3.6. Emissions

Intermittent renewables are the leading technologies for low-carbon power systems. The produced CO₂ emissions (ton of CO₂ equivalent) are calculated based on the amount of energy produced by the Conventional plants according to equation (3).

$$\text{Avoided CO}_2 = P_C(t) \times EF_{\text{Grid}} \quad (4)$$

where $P_C(t)$ is the annual energy generated by conventional resources (TWh) and EF_{Grid} is the grid emission factor. The average emissions from the grid in Egypt is 449.5 g/kWh_{elec} for the base year 2020 [53], based on a fossil fuel energy mix of 84.1 % Natural gas and 15.9 % Oil.

Fig. 27 presents the total equivalent CO₂ emissions for each year, based on the previous results. As the grid accommodates more renewables, equivalent emissions drop, reaching 32.8-million-ton CO₂ equivalent in 2040 from 90 million tons of CO₂ equivalent in 2020.

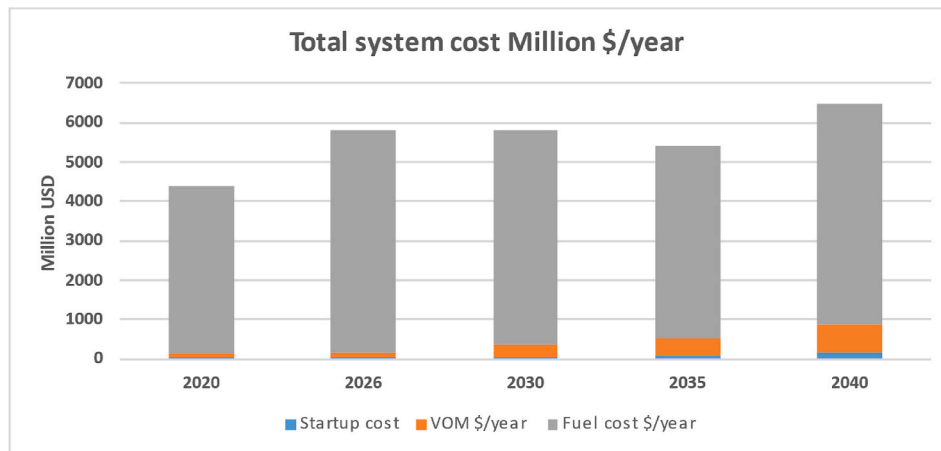


Fig. 23. System cost-breakdown.

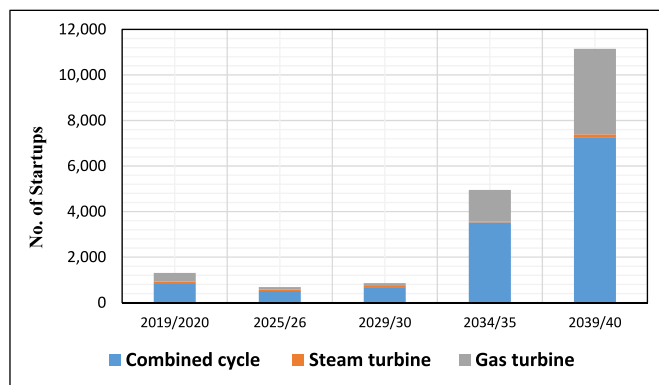


Fig. 24. Number of startups for conventional units.

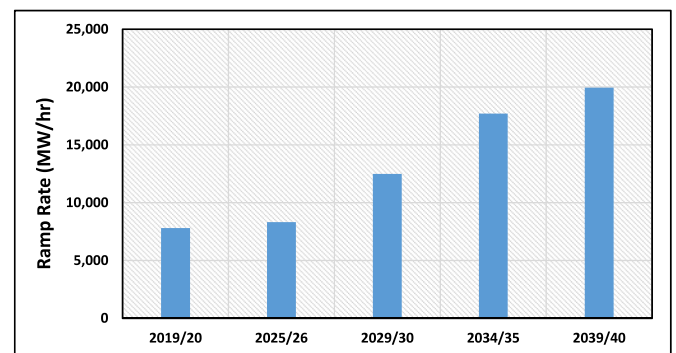


Fig. 26. Maximum system ramp rate of the conventional fleet.

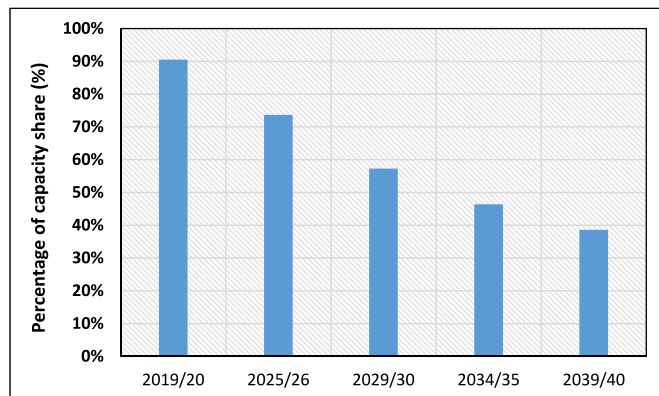
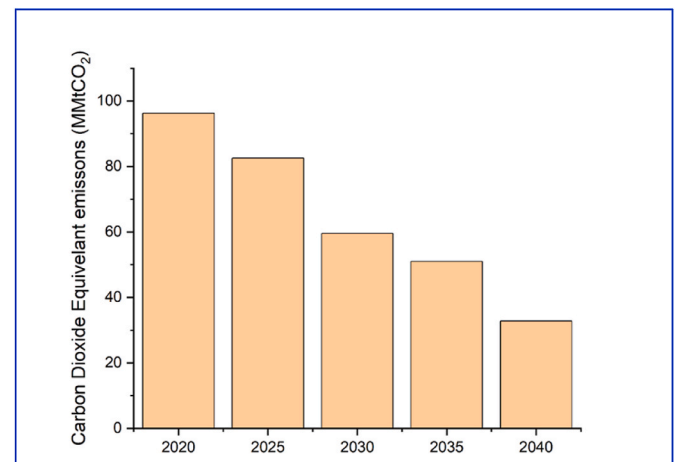


Fig. 25. Capacity share of conventional fleet.

Fig. 27. CO₂ eq emissions' projection.

3.7. Asset management plan

Based on the short-term planning, an asset management plan is proposed to control the usage of thermal fleet units based on their operation schedule and lifetime. The thermal fleet units are classified into five breakdown categories: need for operation for the period from 2020 till 2040, can be retired now due to their expiration, under the efficiency of under capacity, phased out at the first part of the planning period, yet they will be needed in future such that they will be phased in, phased in-phased out-phased in, and phased in-retired before 2040. This asset management plan will help in improving the overall capacity

factor of the conventional fleet, planning repowering, refurbishing or upgrades schedules for phased out power plants that will enter the service again or selling phased out plants in a way that maximizes the system's profit. Fig. 28 shows the asset management plan for the thermal fleet for the whole study period.

4. Conclusions and future work

This study has adopted an approach for discrete and dynamic soft linking between long-term planning, based on the least cost of the

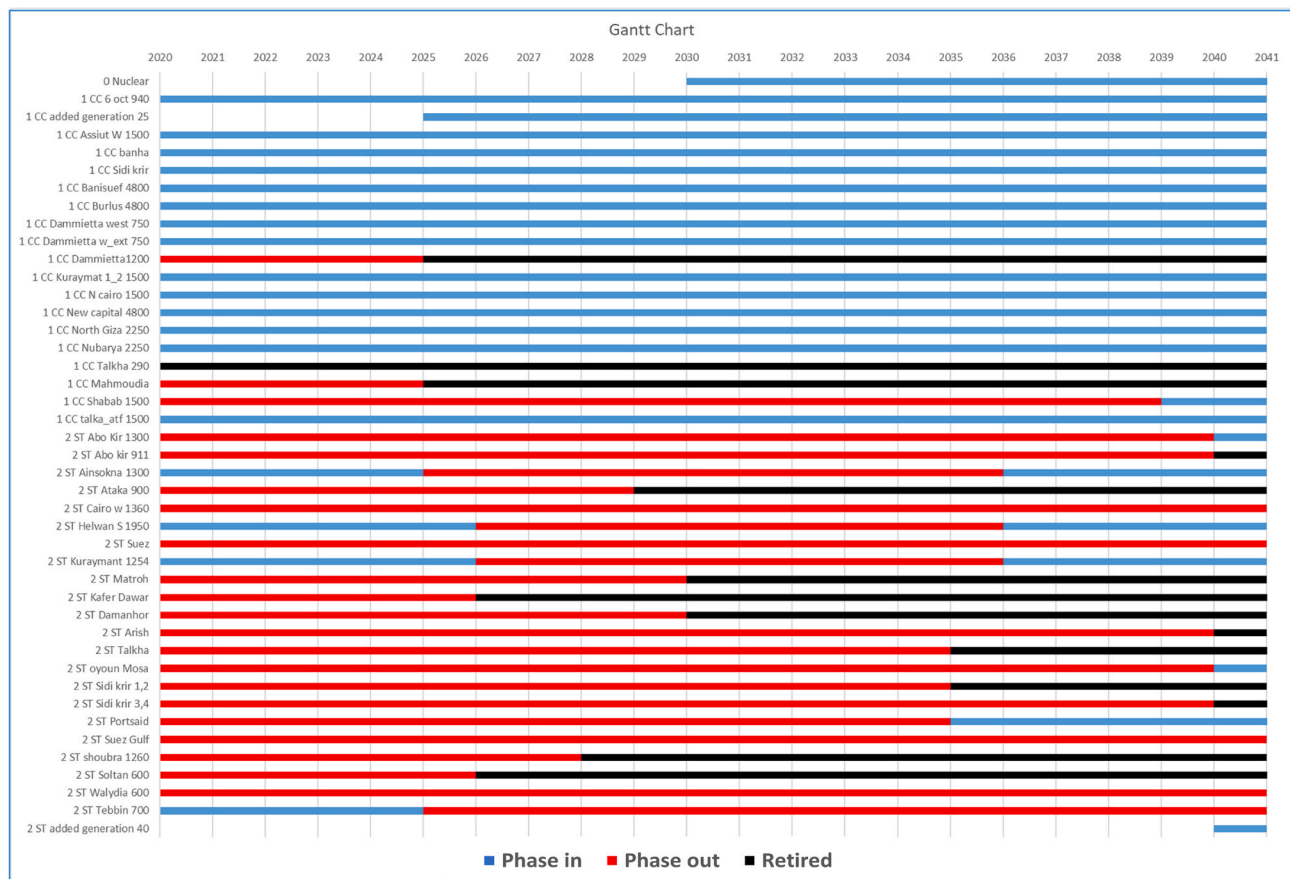


Fig. 28. Asset management plan for the entire conventional fleet.

generated electricity, and short-term analysis, based on system stability and reliability criteria. The long-term planning used the TIMES model, while the short-term analysis used PLEXOS software. This approach was applied to Egypt's under-development energy strategy 2020–2040. The discrete linking is carried out for five milestone years: 2020, 2025, 2030, 2035, and 2040. Applying this approach is necessary to verify that the targeted long-term least cost of electricity produced can be met without a negative impact on the system stability and reliability, which cannot be achieved using long-term planning alone. The results achieved in this work showed that the long-term target could not be achieved on the operational level due to the physical constraints coming from the dispatch characteristics of the conventional units or, as an alternative approach, the need for excessive curtailment of the installed capacity of the RE, such that the least cost cannot be met. This helps in tuning the long-term target such that an optimum solution is achieved, which can ensure the system's reliability and the most possible least cost for the produced electricity. The work has highlighted the importance of renewable energy penetration in reducing the short-run marginal costs associated with the power system. Pumping and storage Hydro plants play an important role in reducing renewables curtailment and increasing system reliability through pumping the water using excess energy and then at night, discharging the water through the turbine; this, in turn, reduces the burden on the required ramp rate during sunset. Egypt-Saudi interconnection represents a reliable solution for exporting excess electricity during the daytime and importing it back during nighttime at zero marginal cost. This helps avoid curtailing excess solar energy during the daytime and use it at night as if the Saudi grid is an additional storage system. Finally, high penetration of renewable energy in the power system will result in reducing about 70 % of the power system emissions compared to 2020, which in turn makes Egypt an active player in achieving the aggressive target of the Paris Agreement of

a 1.5-degree increase compared with that before the industrial revolution by the end of the century.

The proposed methodology can be further enhanced by integrating the grid in the simulation rather than using the assumption of the copper plate for the grid. Also, it introduces the impact of climate change on the variation of wind, solar, and Hydro energy from one year to another. It also considers their stochastic nature rather than considering a deterministic approach and historical approach, such as the case of Hydro energy.

Accordingly, the proposed future work can include.

- The grid can be simulated in zones to determine the impact of power flow among different parts of the grid and check the capacity of the existing interconnections among these zones, keeping in mind that the zones with plenty of renewable generations are not necessarily the zones with the highest demand.
- System inertia analysis can be carried out under some events considering the high share of renewable energy, which affects the system inertia.
- RE resource assessment can be improved by considering P75 exceedance probabilities instead of the P50 values used as a first step. This can be further improved by considering the stochastic nature of renewable energy, specifically wind.
- Finally, the effect of climate change and river Nile water variation need to be included in the performance of hydro dams instead of following a typical historical operation mode.

CRediT authorship contribution statement

Mohamed Hamdi: Writing – original draft, Visualization, Validation, Software, Investigation, Data curation. **Hafez A. El Salmawy:**

Writing – review & editing, Supervision, Project administration, Formal analysis, Conceptualization. **Reda Ragab:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Long-term planning starts with renewable energy penetration by a capacity of 53.2 % during 2040. where the PV capacity is 49.2 GW, the wind capacity of 19.3 GW and the CSP capacity is 4.1 GW. Testing this case on the short-term level led to an imbalance in the power system, and the system was unable to provide the required reserve. Not to mention the required ramp up and down by the conventional fleet, and the excessive cycling of the thermal plants, which reduces its reliability. **Figure A1** shows an example of PLEXOS results obtained in this case. From **Fig. A1** it can be shown that there's a daily cycling for the combined cycle which is technically impossible. Also, steam and gas turbines operate at peak times. This feedback from PLEXOS results in modifying the long-term energy planning, which in turn results in reducing the PV capacity and increasing the wind share, leading to the main topic of the study.

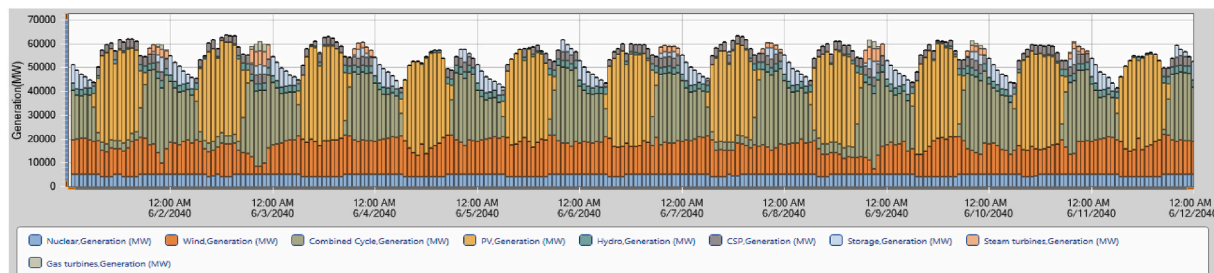


Fig. A1. Short-term validation for the 49.2 GW PV case.

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